

APPENDIX

This appendix provides the complete trial list for the experimental comparison of the DTST actuation system under four configurations: NOPAM, PEA, SEA, and SPEA. The experimental factors were motion frequency, tendon pretension, and PAM pressure, with the pressure level of NOPAM set to 0 psi for consistency in tabulation. Each of the nine parameter conditions was repeated three times for each configuration, resulting in 108 trials in total.

TABLE I

EXPERIMENTAL CONDITIONS BASED ON THE TAGUCHI L9 ORTHOGONAL ARRAY

Run	Configs.	f(Hz)	T_{pre}(N)	P(psi)
1	NO PAM	0.75	30	0
2	NO PAM	0.75	30	0
3	NO PAM	0.75	30	0
4	NO PAM	0.75	45	0
5	NO PAM	0.75	45	0
6	NO PAM	0.75	45	0
7	NO PAM	0.75	60	0
8	NO PAM	0.75	60	0
9	NO PAM	0.75	60	0
10	NO PAM	1	30	0
11	NO PAM	1	30	0
12	NO PAM	1	30	0
13	NO PAM	1	45	0
14	NO PAM	1	45	0
15	NO PAM	1	45	0
16	NO PAM	1	60	0
17	NO PAM	1	60	0
18	NO PAM	1	60	0
19	NO PAM	1.25	30	0
20	NO PAM	1.25	30	0
21	NO PAM	1.25	30	0
22	NO PAM	1.25	45	0
23	NO PAM	1.25	45	0

24	NO PAM	1.25	45	0
25	NO PAM	1.25	60	0
26	NO PAM	1.25	60	0
27	NO PAM	1.25	60	0
28	PEA	0.75	30	5
29	PEA	0.75	30	5
30	PEA	0.75	30	5
31	PEA	0.75	45	13
32	PEA	0.75	45	13
33	PEA	0.75	45	13
34	PEA	0.75	60	20
35	PEA	0.75	60	20
36	PEA	0.75	60	20
37	PEA	1	30	13
38	PEA	1	30	13
39	PEA	1	30	13
40	PEA	1	45	20
41	PEA	1	45	20
42	PEA	1	45	20
43	PEA	1	60	5
44	PEA	1	60	5
45	PEA	1	60	5
46	PEA	1.25	30	20
47	PEA	1.25	30	20
48	PEA	1.25	30	20
49	PEA	1.25	45	5
50	PEA	1.25	45	5
51	PEA	1.25	45	5
52	PEA	1.25	60	13
53	PEA	1.25	60	13
54	PEA	1.25	60	13
55	SEA	0.75	30	5
56	SEA	0.75	30	5
57	SEA	0.75	30	5
58	SEA	0.75	45	13
59	SEA	0.75	45	13
60	SEA	0.75	45	13
61	SEA	0.75	60	20

62	SEA	0.75	60	20
63	SEA	0.75	60	20
64	SEA	1	30	13
65	SEA	1	30	13
66	SEA	1	30	13
67	SEA	1	45	20
68	SEA	1	45	20
69	SEA	1	45	20
70	SEA	1	60	5
71	SEA	1	60	5
72	SEA	1	60	5
73	SEA	1.25	30	20
74	SEA	1.25	30	20
75	SEA	1.25	30	20
76	SEA	1.25	45	5
77	SEA	1.25	45	5
78	SEA	1.25	45	5
79	SEA	1.25	60	13
80	SEA	1.25	60	13
81	SEA	1.25	60	13
82	SPEA	0.75	30	5
83	SPEA	0.75	30	5
84	SPEA	0.75	30	5
85	SPEA	0.75	45	13
86	SPEA	0.75	45	13
87	SPEA	0.75	45	13
88	SPEA	0.75	60	20
89	SPEA	0.75	60	20
90	SPEA	0.75	60	20
91	SPEA	1	30	13
92	SPEA	1	30	13
93	SPEA	1	30	13
94	SPEA	1	45	20
95	SPEA	1	45	20
96	SPEA	1	45	20
97	SPEA	1	60	5
98	SPEA	1	60	5
99	SPEA	1	60	5

100	SPEA	1.25	30	20
101	SPEA	1.25	30	20
102	SPEA	1.25	30	20
103	SPEA	1.25	45	5
104	SPEA	1.25	45	5
105	SPEA	1.25	45	5
106	SPEA	1.25	60	13
107	SPEA	1.25	60	13
108	SPEA	1.25	60	13